Liam Tanner

liamtanner.dev | liam.tanner@brentwood.ca | linkedin.com/liam-tanner | Canadian & Australian Citizen

Technical Skills

Languages: C++, C, Python, SQL, Linux Bash, JavaScript
Cloud & Distributed Systems: AWS (EC2), Linux/UNIX Environments, Relational Databases, Docker, Git
Architecture & Math: Object-Oriented Design, Optimization Math, Algorithms, Data Structures, Concurrency
AI Engineering: PyTorch, LLM Orchestration, Convolutional Neural Networks, Linear/Nonlinear Optimization

Education

University of Victoria — Victoria, BC
Bachelor of Software Engineering — *Sep 2021 – Aug 2026 (expected)*

- 4.00/4.00 GPA - 93% Avg - 2025/2026 Session
- Relevant Coursework: Optimization for Machine Learning, Operating Systems, Databases, Data Mining

Experience

Software Developer — Jun – Dec 2025
Natural Resources Canada — *Victoria, BC*

- Collaborated in an Agile Scrum team to design and implement a scalable ETL data transformation workflow, processing massive relational PostgreSQL datasets and reducing execution times from hours to seconds at scale.
- Engineered an encrypted database release-versioning system using SQL to securely track metadata, ensuring full-lifecycle accountability and seamless feature rollouts across 2-week development sprints.
- Rebuilt legacy frontend infrastructure in JavaScript and wrote custom Linux Bash scripts to automate the deployment process for the entire engineering team.

Software Engineer — Jan 2025 – Aug 2026
AUVIC - Autonomous Underwater Vehicles Club — *Victoria, BC*

- Developed a ROS 2 dynamic configuration pipeline to extract PID and Kalman filter values from YAML files, eliminating C++ recompilation downtime and enabling real-time parameter tuning.
- Resolved critical thruster saturation limits and built a state machine for autonomous navigation, advancing the team to the semi-finals at the international RoboSub 2026 competition.

Research Assistant - Induced Seismicity Project — Sep 2023 – Apr 2024
Natural Resources Canada — *Sidney, BC*

- Contributed to the Induced Seismicity research project by developing complex CO_2 injection simulations and waste-water disposal wells, utilizing Python meshes, and the OpenGeoSys finite element solver; becoming the first NRCan researcher to successfully model a real world site.

Projects

AI Incident Orchestrator | *Python, PostgreSQL, CI/CD, LLM APIs* — May – Aug 2026

- Engineered a production-ready AI orchestration pipeline in a distributed environment, leveraging LLMs and RAG context to automate system incident triage at scale.
- Collaborated in a 17-member team utilizing automated test harnesses and GitHub Actions CI/CD pipelines to validate system resilience and operational quality.

Convolutional Engine | *C++, CUDA* — Apr 2026

- Built a high-performance image processing library from scratch in C++ and CUDA, implementing complex shared memory tiling algorithms to bypass global memory bottlenecks and maximize GPU thread throughput.

Breast Cancer Diagnosis Model | *MATLAB, Optimization Mathematics* — Feb 2026

- Engineered a regularized logistic regression model, applying optimization mathematics (regularized softmax cost function, gradient descent) from scratch to achieve 98.59% accuracy on a complex dataset.

Ray Tracer | *C++, Object-Oriented Design* — Mar – Jun 2025

- Developed a ray tracer capable of rendering 3D animations utilizing core vector math, abstract algorithms, and low-level rendering logic entirely from scratch.

Virtual Triage System | *Python, FastAPI, MongoDB, ReactJS* — Sep – Nov 2024

- Engineered a virtual triaging web application designed to allow prospective hospital, emergency room, patients to wait for an available physician from home.